# Bitmap Indexes

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# Problems

- Ordered Indexes provide good update speeds but…
  - They consume a lot of space
  - Require the database to be sorted
- Hash Indexes provide fast lookup but…
  - Still require a lot of space
  - Good hash index requires good hash function over data

# What Bitmaps Offer

- Much smaller size of index, due to compression
- Fast queries, due to bit operations and size
- They are restricted though…
  - Not made for frequent updates
  - High cardinality of the attributes can be a problem


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# Why use bitmaps?

- Static Databases
  - Lots of databases are static, especially scientific ones
  - BPA data is huge
- Space is cheap, time is not
  - Searching large databases can be time consuming for users


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# What are Bitmap Indexes?

- Bitmaps are often used to represent images
- Although, a bitmap is just an array/matrix of bits
- In this class, when I say Bitmap what I actually mean is Bitmap Index

# Bitmap Creation

- Bitmaps are created buy first observing the possible values that an attribute can take
- Ex. Attribute *Age* may have the range [0,120]
- The possible number of values is called Cardinality
  - Distance might be an attribute with *high* cardinality
  - Graduated might be an attribute with *low* cardinality


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# Bitmap Creation

| Adopted | Species | Age |
|---|---|---|
| No | Dog | 12 |
| Yes | Turtle | 20 |
| Yes | Cat | 1 |
| No | Cat | 4 |

# Bitmap Creation

| Adopted | Species | Age |
|---|---|---|
| No | Dog | 12 |
| Yes | Turtle | 20 |
| Yes | Cat | 1 |
| No | Cat | 4 |

| Yes | No | Dog | Turtle | Cat | 0-10 | 11-20 |
|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 | 1 | 0 |

- Single 1 per tuple per attribute, rest 0s
- Index may even be larger than database, initially


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# Bitmap Creation

| Adopted | Species | Age |
|---|---|---|
| No | Dog | 12 |
| Yes | Turtle | 20 |
| Yes | Cat | 1 |
| No | Cat | 4 |

| Yes | No | Dog | Turtle | Cat | 0-10 | 11-20 |
|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 | 1 | 0 |

- Adopted attribute is binary, so very low cardinality

# Bitmap Creation

| Adopted | Species | Age |
|---|---|---|
| No | Dog | 12 |
| Yes | Turtle | 20 |
| Yes | Cat | 1 |
| No | Cat | 4 |

| Yes | No | Dog | Turtle | Cat | 0-10 | 11-20 |
|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 | 1 | 0 |

- Assuming we only have three species then Species is low cardinality
- What if we don't know how many species?

# Bitmap Creation

| Adopted | Species | Age |
|---|---|---|
| No | Dog | 12 |
| Yes | Turtle | 20 |
| Yes | Cat | 1 |
| No | Cat | 4 |

| Yes | No | Dog | Turtle | Cat | 0-10 | 11-20 |
|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 | 1 | 0 |

- Age is a tricky one, since the range could be large
- Even with max age of 20, lots to consider
  - Do we consider all real numbers?

# Bitmap Creation

| Adopted | Species | Age |
|---|---|---|
| No | Dog | 12 |
| Yes | Turtle | 20 |
| Yes | Cat | 1 |
| No | Cat | 4 |

| Yes | No | Dog | Turtle | Cat | 0-10 | 11-20 |
|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 | 1 | 0 |

- This is where binning comes in
- Age is ‘binned’ into two bins
  - Could be more or less

# Bitmap Creation

| Adopted | Species | Age | Name |
|---|---|---|---|
| No | Dog | 12 | Lexi |
| Yes | Turtle | 20 | Dakota |
| Yes | Cat | 1 | Sky |
| No | Cat | 4 | Zoe |

| Yes | No | Dog | Turtle | Cat | 0-10 | 11-20 |
|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 | 1 | 0 |

- What if we added names?
- How do we make a bitmap of this?

# Bitmap Creation

| Adopted | Species | Age | Name |
|---|---|---|---|
| No | Dog | 12 | Lexi |
| Yes | Turtle | 20 | Dakota |
| Yes | Cat | 1 | Sky |
| No | Cat | 4 | Zoe |

| Yes | No | Dog | Turtle | Cat | 0-10 | 11-20 |
|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 1 | 0 | 0 | 1 | 1 | 0 |

- We can't really…
  - There is no finite number of names,
  - Finite number of species, even though it's large
  - Can create with just the names seen, not ideal though

# There are drawbacks

- Bitmaps excel at databases with low cardinality
  - High cardinality is possible, but some decisions need to be made on either binning or restrictions
- Impose a finite range for attributes with high cardinality or not index over them
- What if we don't index over an attribute?

# Compression

- Remember how the bitmap was larger than data?
  - Not very useful when you want to store the index in memory
- Compression fixes this
  - Uses Run Length Encoding (RLE)
  - WAH, PLWAH, VAL, BBC, lots more…


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# Compression

- Run length encoding is very useful in bitmaps
  - Encodes runs of repeated values as counts of how many
  - Many different ways to count the runs of certain runs
- Bitmaps have many repeated values
  - Since there is only a single 1 for each attribute, lots of 0s

# Compression
# Common Appearances

- Words
  - How big each encoding is, most defined by hardware
- Runs are sequences of either 1s or 0s great than or equal to word size
  - Ex. Word size 8, 00000000 is a run, 00000001 is not
- Dirty bits can ruin compression
  - Words containing dirty bits or a mixture often called literals
- Columns are usually compressed, why?


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# One of the first

- Byte-aligned Bitmap Compression (BBC)
- Developed in 1995, actually a patent out for it
- Very good compression rate
- Complicated rules though…


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# Byte-aligned Bitmap Compression

- Word size is a Byte (hence the name)
- Counts 0s that appear, a byte filled with 0s is run
- A Chunk is encoded gap bytes followed by literals
- Gaps: Runs of 0s
- Literals: bytes that are a mix of 1s and 0s


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# Byte-aligned Bitmap Compression

- Three lengths of runs
  - 0-6: Stored in header byte
  - 7-127: One byte follows header byte counting gaps
  - 128-32767: Two bytes follows header byte counting gaps

# Byte-aligned Bitmap Compression

- Special Bytes:
  - When there is only a single byte that is a literal in the chunk
  - Literal only has a dirty bit
  - Stored as which bit is set
- Otherwise stored as a literal, requires another Byte


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# Byte-aligned Bitmap Compression

- Header Byte:
  - Bits 1-3: Number of runs (the 0-6 size)
  - Bit 4: Indicates whether the tail is special
  - Bits 5-8: Number of literals at end of chunk or dirty bit location

# Byte-aligned Bitmap Compression

- Run Byte(s)
  - If run length is 7-127 bytes or 128-32767 bytes
  - Only occurs if first three bits in header are 111 (filled up)
- Literals
  - Stored exactly as they appear
  - Only occurs when byte is mixture
  - Number is indicated in header byte (bit 4=0)

# Byte-aligned Bitmap Compression Example

**00000000 00000000 00010000 00…00 10101010 11110000**

**13 Bytes**

- Lots to process, so lets look at an example

# Byte-aligned Bitmap Compression Example

**00000000 00000000 00010000 00...00 10101010 11110000**

- Two runs, two bytes
  - Remember word size is 8


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# Byte-aligned Bitmap Compression Example

00000000 00000000 00010000 00...00 10101010 11110000

- Uhoh… dirty bit
  - Following bytes are runs, so looks like we have our first chunk!


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# Byte-aligned Bitmap Compression Example

**00000000 00000000 00010000 00…00 10101010 11110000**

**Compressed Version So Far**

**010**

- Header
  - Bits 1-3: Number of run bytes 010


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# Byte-aligned Bitmap Compression Example

**00000000 00000000 00010000 00...00 10101010 11110000**

**Compressed Version So Far**

**0101**

- Header
  - Bit 4: Is it special?
  - Yes, since there is a dirty bit, so 1


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# Byte-aligned Bitmap Compression Example

00000000 00000000 00010000 00...00 10101010 11110000

**Compressed Version So Far**

01010011

- Header
  - Bits 5-8: Since bit 4 is set, where is the dirty bit?
  - 3rd position, so 0011


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# Byte-aligned Bitmap Compression Example

**00000000 00000000 00010000 00…00 10101010 11110000**

**Compressed Version So Far**

**01010011**

- First chunk
  - Number of runs, Special literal, Dirty bit location


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# Byte-aligned Bitmap Compression Example

**00000000 00000000 00010000 00...00 10101010 11110000**

**Compressed Version So Far**

**01010011**

- Now we have 13 run bytes

# Byte-aligned Bitmap Compression Example

00000000 00000000 00010000 00...00 10101010 11110000

**Compressed Version So Far**

**01010011**

- Followed by a literal
- Next sequence is also a literal
  - Another chunk!


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# Byte-aligned Bitmap Compression Example

**00000000 00000000 00010000 00…00 10101010 11110000**

**Compressed Version So Far**

**01010011 111**

- Header
  - Bits 1-3: 13 run bytes, so more than will fit
  - Set all 3, 111


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# Byte-aligned Bitmap Compression Example

**00000000 00000000 00010000 00…00 10101010 11110000**

**Compressed Version So Far**

**01010011 1110**

- Header
  - Bit 4: Is the literal special?
  - No, so we set to 0


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# Byte-aligned Bitmap Compression Example

**00000000 00000000 00010000 00…00 10101010 11110000**

**Compressed Version So Far**

**01010011 11100010**

- Header
  - Bits 5-8: Since 4 is not set, how many literal bytes?
  - 2 literal bytes, so 0010


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# Byte-aligned Bitmap Compression Example

**00000000 00000000 00010000 00…00 10101010 11110000**

**Compressed Version So Far**

**01010011 11100010 00001101**

- Run Bytes
  - Since we have more than 6 run bytes, need to count them
  - 13 run bytes, so only 1 run byte encoding, 00001101


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# Byte-aligned Bitmap Compression Example

**00000000 00000000 00010000 00…00 10101010 11110000**

**Compressed Version So Far**

**01010011 11100010 00001101 10101010 11110000**

- Literal Bytes
  - We have 2 literals as indicated by the header
  - Encode exactly as they appear, 10101010 11110000


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# Byte-aligned Bitmap Compression Example

**00000000 00000000 00010000 00...00 10101010 11110000**

**Compressed Version So Far**

**01010011 11100010 00001101 10101010 11110000**

- Chunk 1
  - Encoded 24 bits in 8

# Byte-aligned Bitmap Compression Example

00000000 00000000 00010000 00…00 10101010 11110000

**Compressed Version So Far**

01010011 11100010 00001101 10101010 11110000

- Chunk 2
  - Encoded 120 bits in 32 bits


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# Joke Time!

- A programmer is told to “Go to hell!” They find the worst part about that statement is the “Go to”

# Moving on

- Word Aligned Hybrid (WAH)
- Takes up 50% more space compared to BBC
- Performs queries 12 times faster on practical and synthetic databases!
- Much much simpler than BBC

# Word Aligned Hybrid

- Uses word size based on hardware
  - 32, 64, someday maybe even 128 bits!
  - Can define any word size though
  - Why is using word size based on hardware a good idea?
- If word size is 32, then encodes runs/literals of 31

# Word Aligned Hybrid

- Header is actually one 1 or two bits
- Run
  - 1000000000000010 = 0000000000000000 0000000000000000
  - 1100000000000010 = 1111111111111111 1111111111111111
- Literal
  - 0000011000000010 = 00001100000010


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# Word Aligned Hybrid Example

**00000000 00000000 00010000 00...00 10101010 11110000**

**13 Bytes**
**104 bits**

- Same bit string as before, only this time we will use 32 bit words


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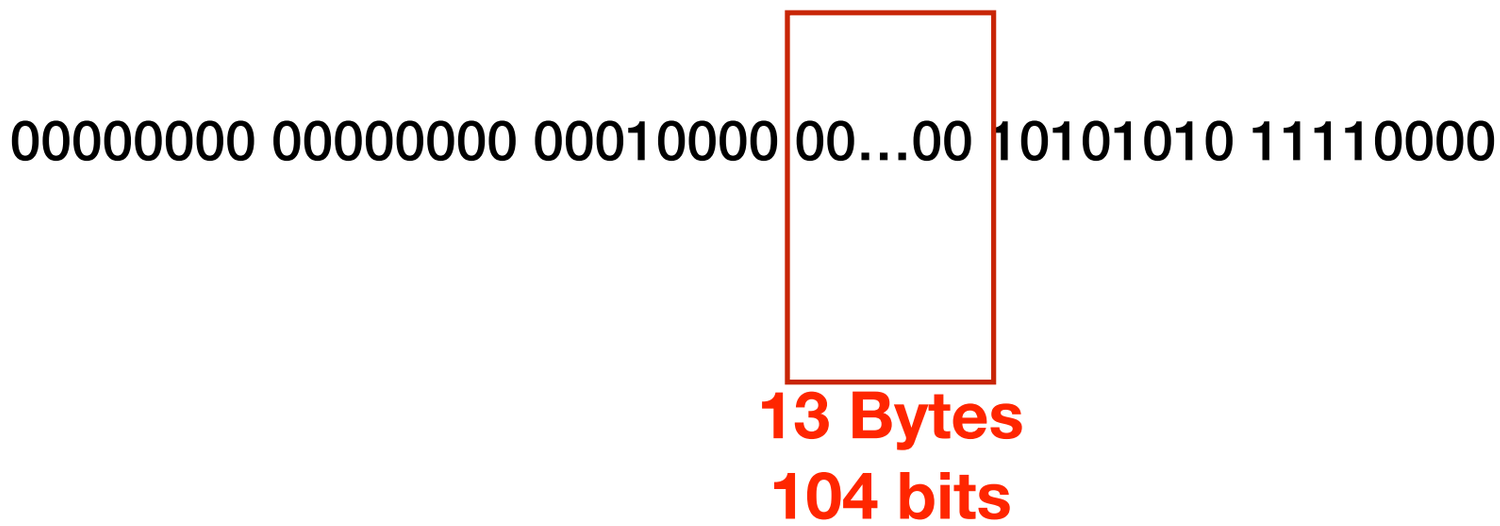
# Word Aligned Hybrid Example




- Look at 31 bit sections and process them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# Word Aligned Hybrid Example

**00000000 00000000 00010000 00...00 10101010 11110000**

**Compressed Version of Selected**

**00000000 00000000 00010000 00000000**

- First 31 bits is a mix, so store as a literal

# Word Aligned Hybrid Example

00000000 00000000 00010000 00...00 10101010 11110000

**13 Bytes - 7 bits**
**97 bits**

- Since we took some out of here, only 97 bits now


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# Word Aligned Hybrid Example

**97 - 31 = 66 bits left**

00000000 00000000 00010000 00...00 10101010 11110000

**Compressed Version of Selected**

**10000000 00000000 00000000 00000001**

- Now onto the next 31 bits
- All 0s so we start a run


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# Word Aligned Hybrid Example

**66 - 31 = 35 bits left**

00000000 00000000 00010000 00...00 10101010 11110000

**Compressed Version of Selected**

**10000000 00000000 00000000 00000010**

- Now onto the next 31 bits
- Still only 0s, so we increment the count of our run


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# Word Aligned Hybrid Example

**35 - 31 = 4 bits left**

00000000 00000000 00010000 00...00 10101010 11110000

**Compressed Version of Selected**

**10000000 00000000 00000000 00000011**

- Now onto the next 31 bits
- Still only 0s, so we increment the count of our run, again


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# Word Aligned Hybrid Example

**35 - 31 = 4 bits left**

00000000 00000000 00010000 00…00 10101010 11110000

**Compressed Version of Selected**

- There are only 12 bits left
  - How do we store these now?


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# Word Aligned Hybrid Example

**35 - 31 = 4 bits left**

00000000 00000000 00010000 00...00 10101010 11110000

**Compressed Version of Selected**

000000 10101010 11110000

- Since all the columns have same number of tuples
  - We can just encode it as a shorter literal


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# Variable Aligned Length

- Variable Aligned Length (VAL)
- Fixes some of the shortcomings of WAH
  - WAH is susceptible to dirty bits, why?
- I worked on this one a little during my M.S.
- Very similar to WAH, so we will be brief

# Variable Aligned Length

- VAL introduces Segments into words
  - Basically miniature words inside words
- 16 bit word, 7 bit segments
  - 01 1010001 1000010 = 1010001 1111111 1111111

# Variable Aligned Length

- VAL introduces Segments into words
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# Variable Aligned Length

- VAL introduces Segments into words
  - Basically miniature words inside words
- 16 bit word, 7 bit segments
  - 01 1010001 1000010 = 1010001 1111111 1111111

# Variable Aligned Length

- VAL introduces Segments into words
  - Basically miniature words inside words
- 16 bit word, 7 bit segments
  - 01 1010001 1000010 = 1010001 1111111 1111111
- 32 bit word, 7 bit segments, 4 segments
- 32 bit word, 15 bit segments, 2 segments

# Variable Aligned Length

- VAL tries to improve on WAHs compression
- Same query times as WAH in most cases
- Smaller segment sizes mean better compression
  - To a point…
  - Why do you think that is?

# Position List Word Aligned Hybrid

- Position List Word Aligned Hybrid (PLWAH)
- Begins just like WAH
  - We start the compression the same
  - Adds an additional step when WAH would normally completes
- Combines “Near Identical” words with fill words
  - Nearly identical means having the same fill type besides the dirty bit


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# Position List Word Aligned Hybrid

1 0000000000000000000000000000001
0 0000000000000000001000000000000
1 0000000000000000000000000000010
0 0000000100000000000000000000000
0 0000000000001000000000000000000

- Some compressed words using WAH


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# Position List Word Aligned Hybrid

1 00000000000000000000000000000001
0 00000000000000000001000000000000
1 00000000000000000000000000000010
0 00000001000000000000000000000000
0 00000000000001000000000000000000

- These are fills


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# Position List Word Aligned Hybrid

1 00000000000000000000000000000001
0 00000000000000000001000000000000
1 00000000000000000000000000000010
0 00000001000000000000000000000000
0 00000000000001000000000000000000

- These are literals

# Position List Word Aligned Hybrid

1 0 00000 00000000000000000000000001
0 00000000000000000001000000000000
1 0 00000 00000000000000000000000010
0 00000001000000000000000000000000
0 00000000000001000000000000000000

- PLWAH uses the 5 bits after the first two
  - Signifies where the dirty bit is


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# Position List Word Aligned Hybrid

1 0 00000 00000000000000000000000001
0 00000000000000000001000000000000
1 0 00000 00000000000000000000000010
0 00000001000000000000000000000000
0 00000000000001000000000000000000

- Using these, the literals following a fill can be combined into the fill


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# Position List Word Aligned Hybrid

1 0 00000 00000000000000000000000001
0 00000000000000000011000000000000
1 0 00000 00000000000000000000000010
0 00000001000000000000000000000000
0 00000000000010000000000000000000

- Dirty bit is 20 in


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# Position List Word Aligned Hybrid

1 0 10100 00000000000000000000000001
1 0 00000 00000000000000000000000010
0 00000001000000000000000000000000
0 00000000000010000000000000000000

- Combined with the fill before, we set the “nearly identical” value to the position of the dirty bit
  - Dirty bit was in position 20


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# Position List Word Aligned Hybrid

1 0 10100 00000000000000000000000001
1 0 00000 00000000000000000000000010
0 00000001 00000000000000000000000
0 00000000000001 00000000000000000

- Do the same for the next fill and literal
  - Dirty bit is in position 8


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# Position List Word Aligned Hybrid

1 0 10100 00000000000000000000000001
1 0 01000 00000000000000000000000010
0 00000000000010000000000000000000

- Set the “nearly identical” bits to the position

# Position List Word Aligned Hybrid

1 0 10100 00000000000000000000000001
1 0 01000 00000000000000000000000010
0 00000000000010000000000000000000

- Notice, only the fill words have the 5 bits for the position of the dirty bit
- Literals are still encoded with 31 bits and 1 bit to signify that word is a literal


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# Position List Word Aligned Hybrid

- There are algorithms to define the size of the “nearly identical” bits
  - Not covered in this class, but I encourage those interested to read the paper

# Queries

- Bitmap Querying happens between columns
  - Really fast since all the operations are bit operations
  - Select columns that you want to compare
- Range and Point queries
  - AND handles point
  - OR handles range

# Queries

- Bit operations between literals are simple
- Bit operations between compressed runs are still simple
  - Take the minimum run size
  - Then perform the bit operation on the two fill numbers

# Queries

**Want to find all Dogs that are Adopted**
**1860 bits/tuples**

| | | | | |
|---|---|---|---|---|
| Dog: | 01010101..01 | 1000..0111010 | 011111111..11 | |
| Adopted: | 01111111..11 | 010101010..01 | 1000..0111001 | 011111111..11 |

# Queries

Dog & Adopted

| | | | | |
|---|---|---|---|---|
| Dog: | 01010101..01 | 1000..0111010 | 011111111..11 | |
| | & | | | |
| Adopted: | 01111111..11 | 010101010..01 | 1000..0111001 | 011111111..11 |

Result: 010101010..01

# Queries

**26 runs = 806 bits** Dog & Adopted

| | | | | |
|---|---|---|---|---|
| **Dog:** | **01010101..01** | **1000..0111010** | **011111111..11** | |
| | | & | | |
| **Adopted:** | **01111111..11** | **010101010..01** | **1000..0111001** | **011111111..11** |
| Result: | 010101010..01 | 000000000..00 | | |

# Queries

25 runs = 775 bits Dog & Adopted

| | | | |
|---|---|---|---|
| **Dog:** | **01010101..01** | **1000..0111010** | **011111111..11** |
| | | & | |
| **Adopted:** | **01111111..11** | **010101010..01** | **1000..0111001** | **011111111..11** |
| Result: | 010101010..01 | 000000000..00 | 1000..0111001 |

# Queries

Dog & Adopted

| | | | | |
|---|---|---|---|---|
| **Dog:** | **01010101..01** | **1000..0111010** | **011111111..11** | |
| | | | & | |
| **Adopted:** | **01111111..11** | **010101010..01** | **1000..0111001** | **011111111..11** |
| | | | ↓ | |
| Result: | 010101010..01 | 000000000..00 | 1000..0111001 | 011111111..11 |

# Queries

Dog & Adopted

| | | | | |
|---|---|---|---|---|
| **Dog:** | **01010101..01** | **1000..0111010** | **011111111..11** | |
| **Adopted:** | **01111111..11** | **010101010..01** | **1000..0111001** | **011111111..11** |
| Result: | 010101010..01 | 000000000..00 | 1000..0111001 | 011111111..11 |

# Sorting

- Sorting bitmap indexes can drastically improve compression ratios
- Grey Code sorting can be performed on a bitmap
- Lexicographical sorting on the data itself is often just as effective
- But wait… didn't we say that one of the pros of Bitmap Indexing was that sorting wasn't required?

# Compression Time!

- WAH vs. VAL
- 28 ones, 28 zeros, 28 ones, 28 zeros
- Compress using 32-bit words
  - VAL: 15 bit segments and 7 bit segments